

community because performing experiments on them is both an expensive and time-consuming process.

There are a number of simulators that facilitate the evaluation of various aspects of IoT systems. Many of these are available for cloud computing and some of them are used to simulate the fusion of edge computing in an IoT environment. Here we briefly describe well-known edge computing simulators used for modelling, and compare them in terms of quality. Figure 3 shows the various simulators/emulators available for edge computing. Table 1 gives a comparison between these.

iFogSim simulator is used for simulating fog and edge computing, and to provide an evaluation tool to manage various resources. It is designed based on the CloudSim simulator, which is primarily used to simulate a cloud computing environment. It provides a quantitative evaluation of latency, energy utilisation, and network traffic.

IoTSim simulates edge computing scenarios, where a huge amount of data is processed in a Big Data processing system by the IoT system. It gives special focus on simulating communication and network delays among data storage and data processing virtual machines.

Table 1: A comparison of edge simulators/emulators

Attributes	iFogSim	IoTSim	Edge CloudSim	FogNet-Sim++	EmuFog	FogBed
Scalability	×	√	×	√	×	×
Mobility	×	×	×	√	×	×
Network delay	×	×	√	√	√	×
Energy consumption	√	×	×	√	×	×
Bandwidth utilisation	√	√	√	×	×	×

EdgeCloudSim is explicitly designed for the evaluation of edge computing scenarios in IoT. It provides various models such as the mobility module, edge server module, and network link module to assess the several features of edge computing. It is user-friendly and provides a method to configure devices and applications using XML files.

FogNetSim++ is based on the prominent simulator OMNET++, which contains widespread network libraries and provides numerous network protocols. It empowers researchers to integrate user-defined mobility models and manage delivery mechanisms.

EmuFog emulator supports repeatable and manageable experiments with real-life applications. It emulates network topologies and is designed on

top of the MaxiNet network emulator. However, it does not support mobility, both for fog nodes and clients, and also doesn't provide hierarchical fog infrastructures.

Fogbed is an extension of the Mininet simulator. The network topology can be changed dynamically using Docker containers. However, it does not support the main facets of fog computing such as fault tolerance, scalability, security, and reliability.

IoT devices must be able to process data closer to end users, especially for time-critical applications. Edge computing is the best option for this. However, it's still in an early stage of development and more research is needed for the efficient integration of edge computing with IoT. **END** 🐧

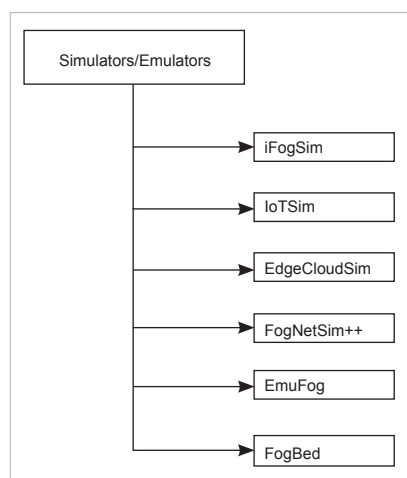


Figure 3: Simulators/emulators for edge computing

References

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